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Effect of infill structure on mechanical properties of 3D printed components

P Košťál[1*], D.R. Delgado Sobrino[1], R. Holubek[1], M. Matúšová[1]

[1] Slovak University of Technology in Bratislava, Materials Sciences and Technology in Trnava, Institute of Production Technologies, 91701 Trnava, Slovakia

*E-mail: peter.kostal@stuba.sk

Abstract. This paper deals with experimental determination of the influence of internal infill geometry and its density on the mechanical properties of 3D printed components made from thermoplastic PET-G material. Tensile tests were performed on 33 specimens prepared according to DIN EN ISO 527 standard, testing five standard infill structures (rectilinear, honeycomb, gyroid, cubic, line) at densities of 15% and 30%. Specimens were manufactured using FDM technology with constant printing parameters on an Instron 3382 testing machine. Results showed that honeycomb structure at 30% density achieved the highest tensile strength of 36.8 MPa, while the transition from 15% to 30% density brought an increase in strength of 85-102% for all tested geometries. Gyroid structure exhibited the most balanced ratio of strength (33.1 MPa) and deformability (5.50%), making it suitable for cyclic loading applications. Fractographic analysis confirmed interlaminar delamination as the primary failure mechanism. The study demonstrated that selection of optimal internal structure significantly affects mechanical properties of 3D printed parts, with honeycomb or cubic structures at minimum 30% density recommended for applications requiring high tensile strength.

1. Introduction

Additive manufacturing has established itself as a key technology of modern engineering, transcending the boundaries of prototyping and entering serial production of functional components [1]. The efficiency of these processes is no longer tied exclusively to time and cost savings, but primarily to the reliability and predictability of mechanical properties of final products [2].

Additive manufacturing technology has undergone dynamic development since its inception in 1986, when Charles Hull patented stereolithography (SLA) and simultaneously introduced the universal STL format [3]. The original SLA technology involved gradual curing of photosensitive polymer resin with a laser beam. In the following decades, other key technologies were established: FDM (Fused Deposition Modeling) based on melting thermoplastic filaments [4], SLS (Selective Laser Sintering) utilizing sintering of powder materials [5], MJF (Multi Jet Fusion) with precise application of binding agents [6], and PolyJet technology working with liquid resins [7].

With the diversification of 3D printing processes, studies gradually emerged demonstrating the fundamental influence of internal structure (infill) on final product properties [8]. Internal infill affects not only weight and printing time, but primarily the mechanical characteristics of parts [9]. ISO/ASTM 52921 standard defines methodology for static tensile and compression tests for components manufactured by additive techniques [10].

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From economic and ecological perspectives, infill density optimization represents significant potential for material and energy savings [11]. Geometries such as honeycomb or gyroid are considered advantageous in terms of uniform stress distribution and efficient material utilization [12, 13].

Despite extensive research on standard structures, work focused on industrial applicability of these findings and their impact on manufacturing process reliability is still lacking [14]. This paper attempts to fill this gap through systematic comparison of five internal infill geometries at two density levels via tensile testing.

2. Experimental methodology

2.1 Specimen preparation

Test specimen geometry was designed in accordance with DIN EN ISO 527-1 standard [15], which defines standard dimensions and shape of specimens for tensile tests of thermoplastics. A three-dimensional model of test bar with gauge length of 75 mm, width of 10 mm and thickness of 4 mm was created in Onshape software.

Before printing, all specimens were oriented identically - gauge axes parallel to printing direction (0° orientation), eliminating the influence of material anisotropy on result comparability, Figure 1.

The CAD model was subsequently processed in PrusaSlicer 2.4.2 software with the following printing parameters:

- Layer height: 0.2 mm
- Nozzle temperature: 240 °C
- Bed temperature: 85 °C
- Print speed: 60 mm/s
- Number of perimeter layers: 3
- Material: PET-G (polyethylene terephthalate glycol)

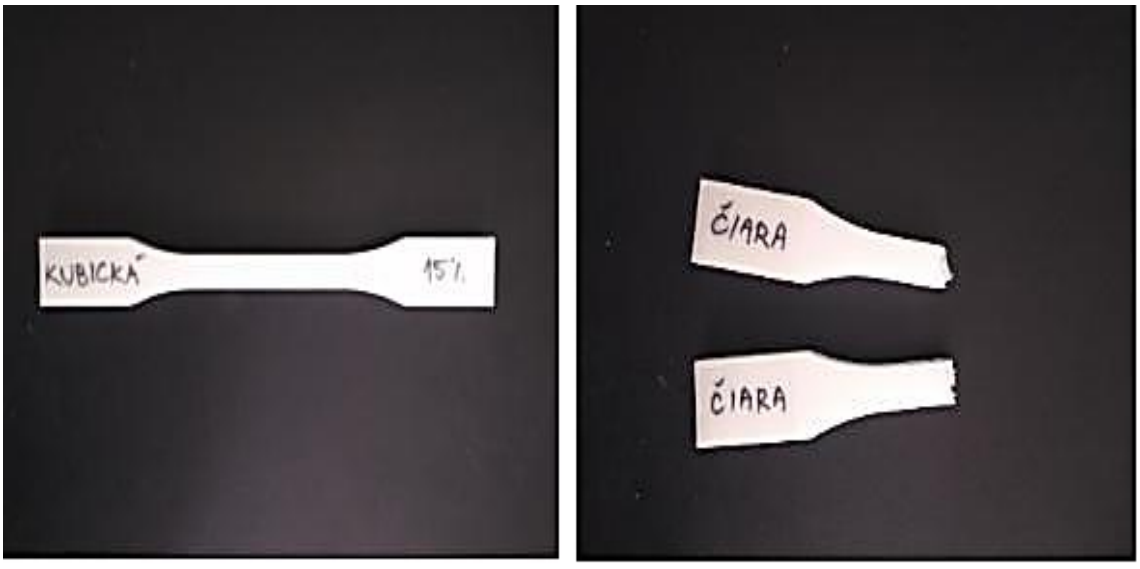


Figure 1. Printed test specimen before and after tensile test

2.2 Internal infill structures

Five different internal infill geometries were defined in PrusaSlicer:

1. **Rectilinear** - parallel lines alternately oriented at ±45°

2. **Honeycomb** - hexagonal cells with optimal stiffness
3. **Gyroid** - triperiodic minimal surface without sharp edges
4. **Cubic** - three-dimensional lattice structure
5. **Line** - unidirectional parallel lines

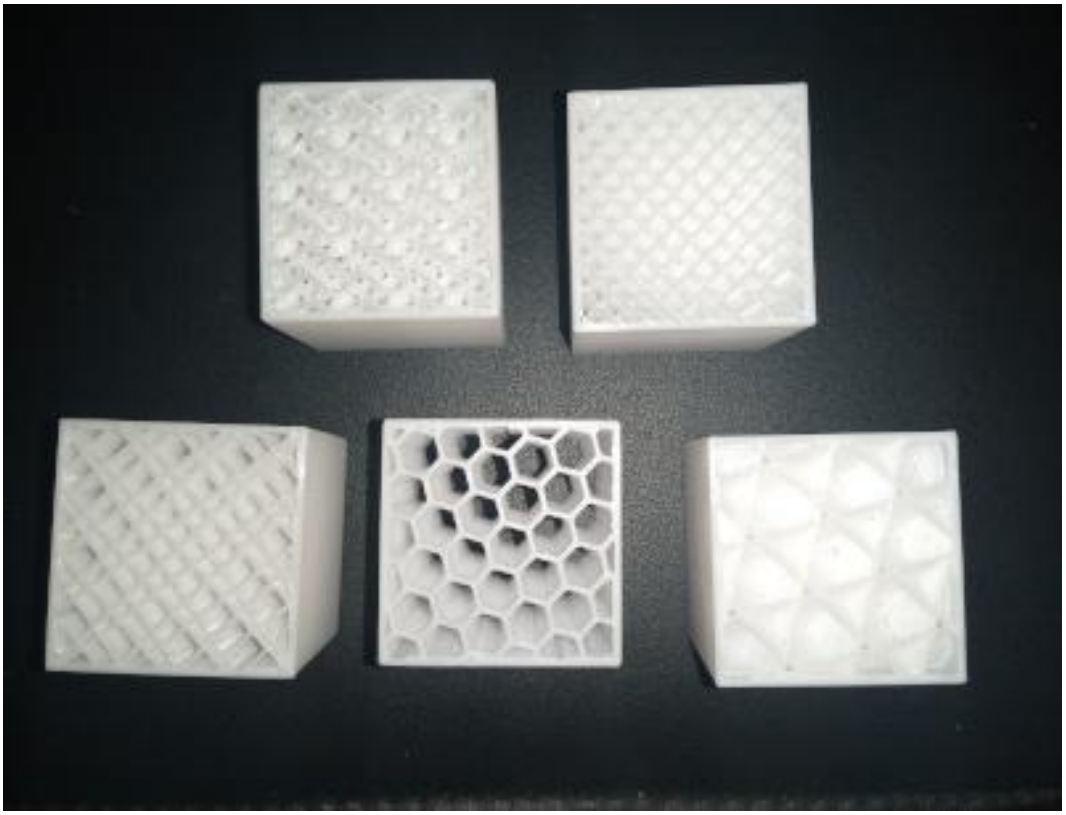


Figure 2. Examples of used infill structures

Two sets of specimens with internal density of 15% and 30% were prepared for each structure. Additionally, a third series at 100% density was manufactured for rectilinear structure as reference value. Total number of specimens: 5 structures × 2 densities × 3 replicas + 1 structure × 1 density × 3 replicas = 33 specimens, Figure 2.

2.3 Tensile testing

Mechanical tests were performed on universal testing machine Instron 3382 with 100 kN capacity, Figure 3.

Experimental protocol included:

- Specimen clamping in pneumatic grips with gauge length of 50 mm
- Application of initial preload of 5 N
- Test initiation at constant rate of 5 mm/min
- Continuous recording of force and displacement at 10 Hz frequency

Automatic test termination at 40% force drop

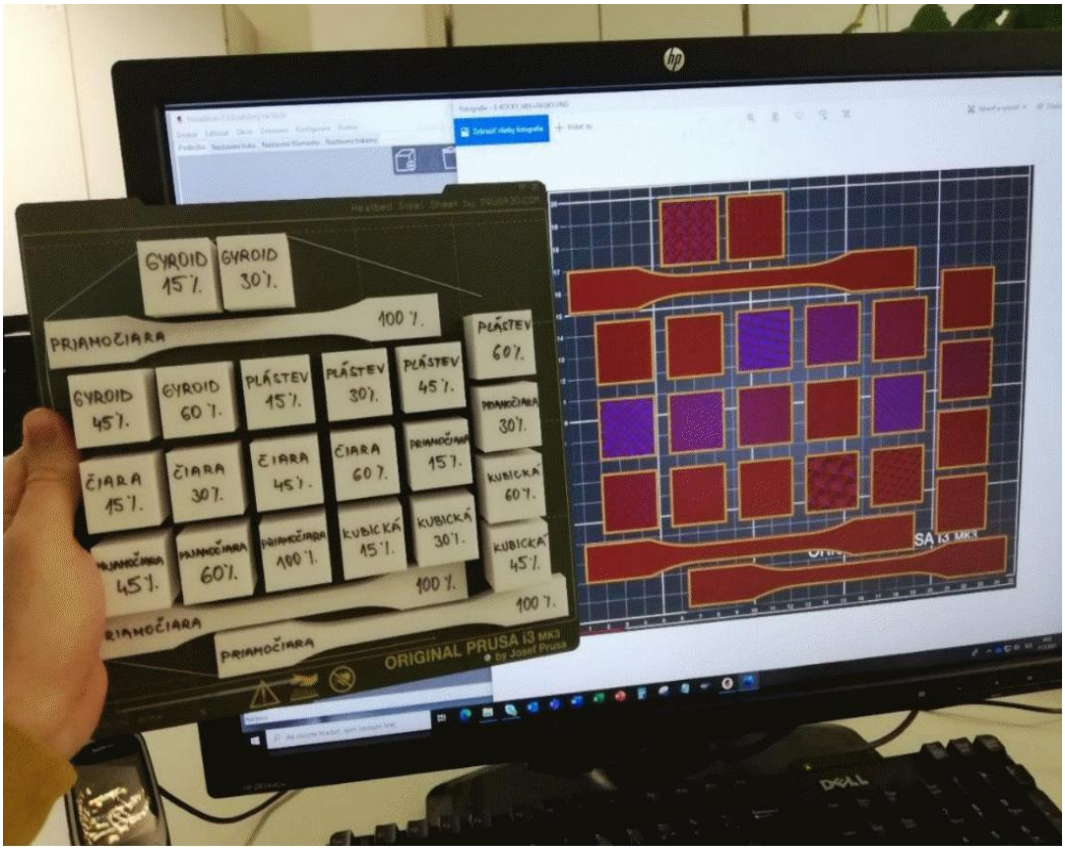


Figure 3. Examples of used infill structures

2.4 Data processing

Measured data were exported to MATLAB R2021b environment where basic mechanical characteristics were calculated according to standard relationships. Raw data from Instron 3382 testing apparatus were stored in .csv format with 10 Hz sampling frequency. Each file contained time stamps, force [N] and displacement [mm] values.

Prior to actual processing, data quality control was performed, identifying and removing invalid values arising during test initiation and termination. This phase included removal of initial fluctuations during specimen settling and final oscillations after fracture.

A low-pass filter with 5 Hz cutoff frequency was applied to suppress noise in measured signals. This eliminated high-frequency oscillations caused by testing machine vibrations without loss of essential information about specimen mechanical behaviour.

Stress: $$\sigma = \mathrm{F}/\mathrm{A}_0 \quad (1)$$

Relative elongation: $$\varepsilon = \Delta L / L_0 = (L - L_0) / L_0 \quad (2)$$

Elastic modulus: $$E = \sigma / \varepsilon = \Delta\sigma / \Delta\varepsilon \quad (3)$$

where F is instantaneous force [N], A_0 is initial specimen cross-section [mm²], L_0 is initial gauge length [mm] and L is instantaneous length [mm].

3. Results

3.1 Stress-strain characteristics

Analysis of stress-strain curves revealed characteristic behaviours for individual infill types. Specimens with honeycomb at 30% density exhibited pronounced linear elasticity up to ultimate tensile strength, followed by brittle fracture without significant plastic deformation, Figure 4.

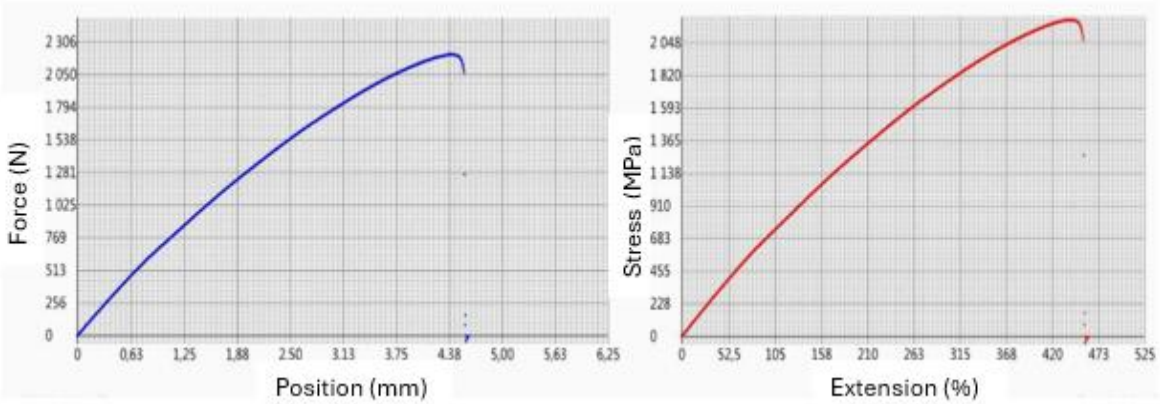


Figure 4. Examples of used infill structures

3.2 Quantification of mechanical properties

Shapiro-Wilk test confirmed normal distribution of all measured values ($p > 0.05$), validating the use of parametric statistical tests, Figure 5.

Table 1. Mechanical properties of specimens at 30% infill density

Structure	Rm [MPa]	σ [MPa]	εf [%]	σ [%]
Honeycomb	36.8	0.9	4.56	0.16
Cubic	35.8	1.0	5.40	0.21
Gyroid	33.1	1.3	5.50	0.24
Rectilinear	31.0	1.0	5.91	0.24
Line	31.8	1.4	5.74	0.29

3.3 Effect of infill density

Table 2. Effect of infill density on tensile strength (*Student's t-test)

Structure	15% [MPa]	30% [MPa]	Increase [%]	p-value
Honeycomb	18.2 ± 0.8	36.8 ± 0.9	+102	< 0.001
Rectilinear	16.8 ± 1.2	31.0 ± 1.0	+85	< 0.001
Gyroid	17.5 ± 1.1	33.1 ± 1.3	+89	< 0.001

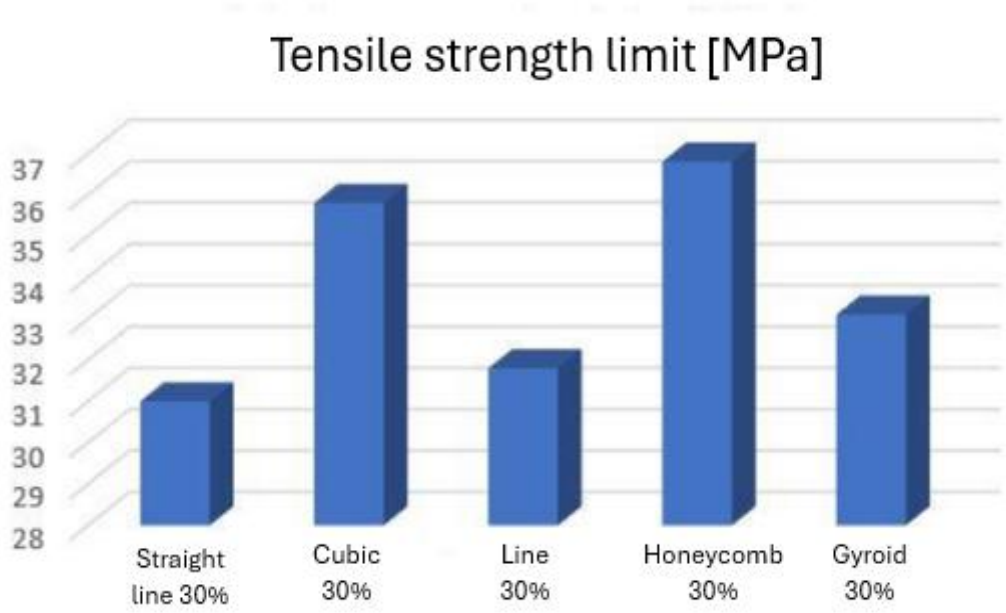


Figure 5. Comparison of tensile strength for different infill types at 30%

3.4 Failure mechanism characterization

For deeper understanding of failure mechanisms, detailed fractographic analysis of fracture surfaces was performed on representative specimens with honeycomb infill at 30% density using optical microscopy at various magnifications, Figure 6.

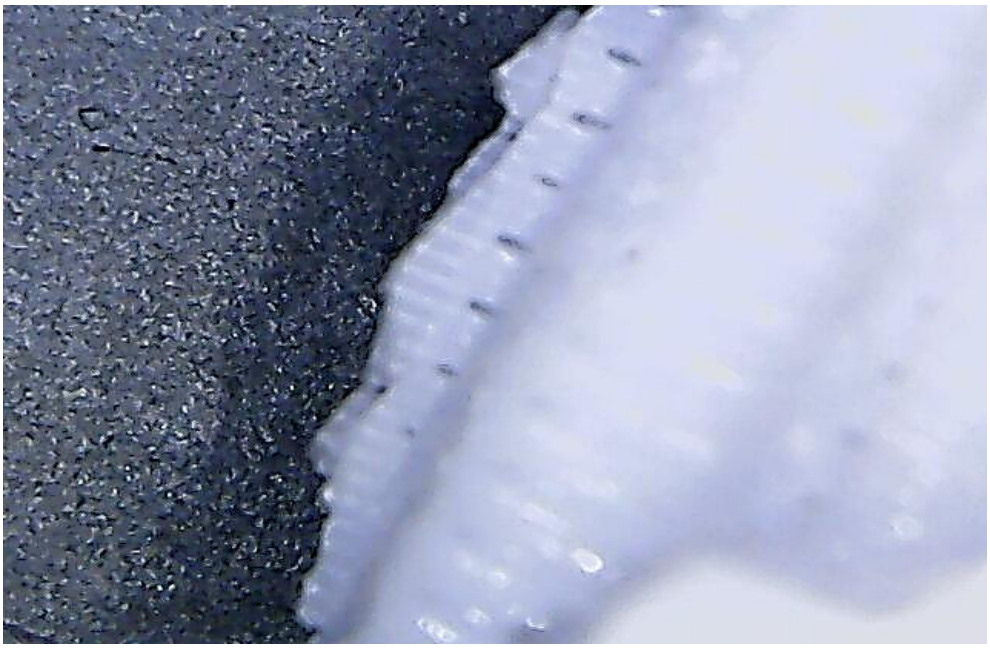


Figure 6. Macroscopic view of fracture surface of PET-G specimen with honeycomb infill at 30% density

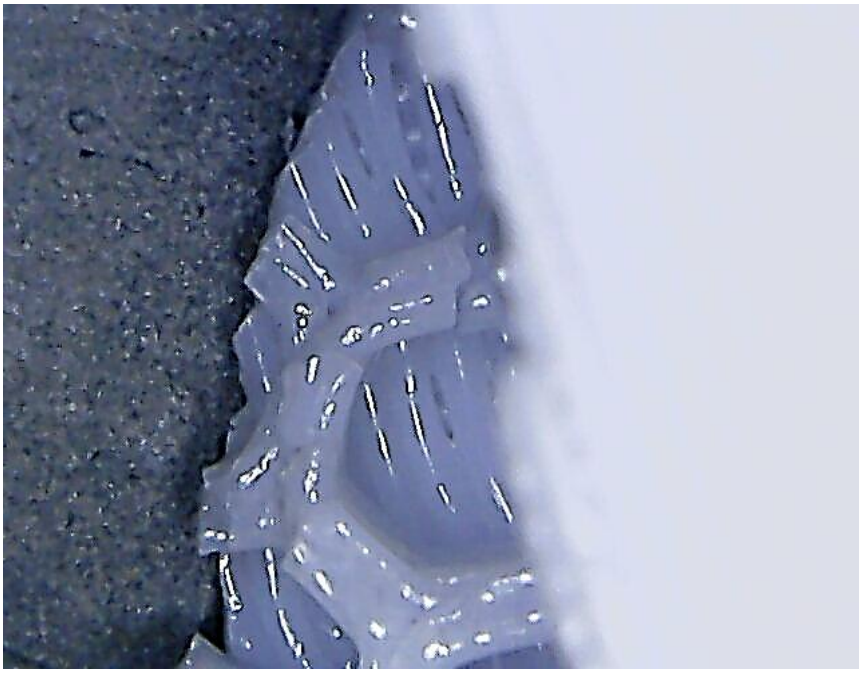


Figure 7. Detail of fracture surface at higher magnification showing layered structure and failure mechanisms

Identified failure mechanisms:

- Interlaminar delamination: Clearly visible separation of individual printing layers, typical for FDM technology under tensile loading.
- Cohesive failure of PET-G matrix: Visible plastic deformations with characteristic fibrils indicating ductile nature of base polymer.
- Honeycomb infill structure failure: Localized failure at stress concentration points, primarily at hexagonal cell nodes Figure 7.

4. Discussion

Results confirm the fundamental influence of internal infill geometry on mechanical properties of 3D printed components. Honeycomb structure achieves highest tensile strength due to efficient stress distribution through hexagonal cells that minimize local stress concentrations [16-21].

Fractographic analysis confirmed theoretical assumptions about layered structure influence on FDM printed part mechanical properties. Observed interlaminar delamination corresponds with findings published in literature [22-28], where authors identified insufficient inter-layer adhesion as primary limiting factor for strength in direction perpendicular to printing layers.

For applications requiring high reliability under tensile loading (e.g., tool holder brackets, protective covers for electronic devices), honeycomb or cubic structures at minimum 30% density are recommended.

Economic analysis shows that at material costs of 0.03 €/g for PET-G filament, increasing density from 15% to 30% represents additional costs of 0.42 € per specimen, which with strength increase of 85-102% represents cost-effective solution.

5. Conclusion

Based on systematic tensile tests of 33 specimens with five standard internal infills at two densities, the following conclusions can be formulated:

- **Honeycomb structure at 30% density** achieves highest tensile strength (36.8 ± 0.9 MPa) with lowest result variability.
- **Cubic structure** represents practically equivalent alternative (35.8 ± 1.0 MPa) with slightly higher deformability (5.40%).
- **Gyroid geometry** provides optimal compromise between strength (33.1 ± 1.3 MPa) and flexibility (5.50%).
- **Infill density** represents critical parameter - transition from 15% to 30% doubles tensile strength for all tested geometries.

Fractographic analysis confirmed interlaminar delamination as primary failure mechanism.

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